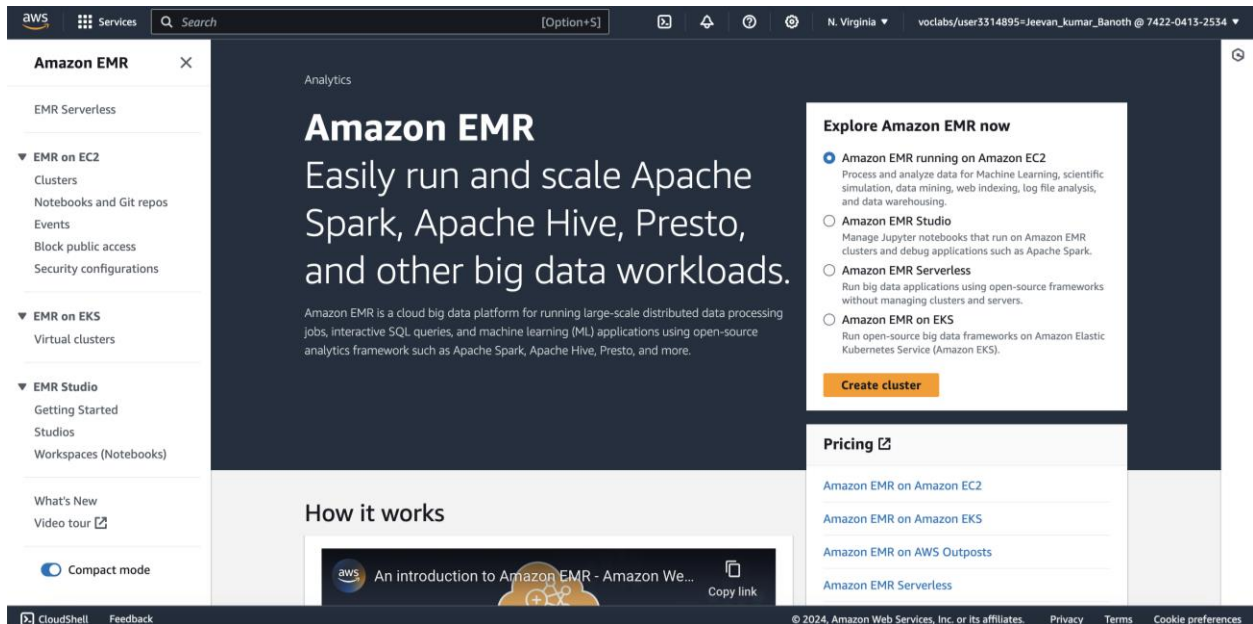


Jeevan Kumar Banoth - 02105145
CIS 602-7103: Special Topics in CIS

Homework – 5:

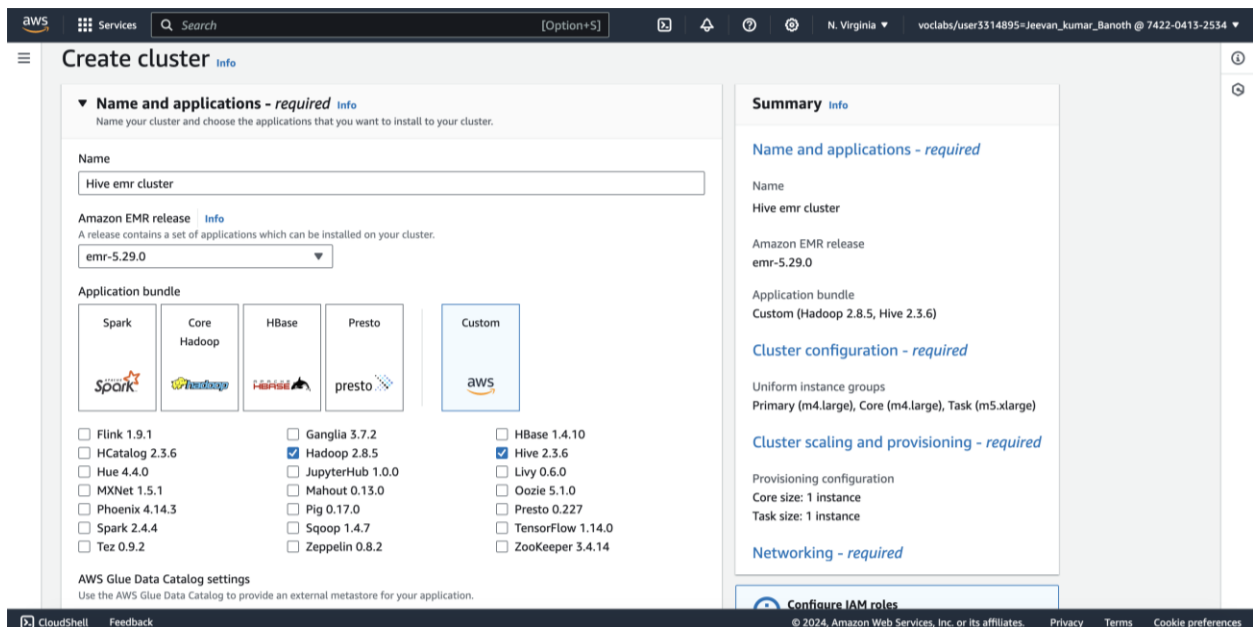
Task 1: Install an Amazon EMR Cluster.

To find EMR, use the Services search box in the AWS Management Console. Look out for the EMR entry on the list.



Choose "Create cluster" from the menu.

Click the Go to advanced choices button located in the upper right corner of the Create Cluster - Quick choices screen.



In Software Configuration, choose the Release option emr-5.29.0.

Verify that the following programs are selected:

Hadoop 2.8.5 2.3.6 Hive

Delete Hue and Pig as well as any other selected apps.

Leaving everything as it is, click Next.

The screenshot shows the AWS EMR console's cluster configuration page. The 'Cluster configuration - required' section is active, showing options for 'Uniform instance groups' (selected) and 'Flexible instance fleets'. Under 'Uniform instance groups', the 'Primary' group is configured with 'm4.large' instance type. The 'Core' group is also configured with 'm4.large' instance type. The 'Summary' panel on the right shows the cluster name 'Hive emr cluster', Amazon EMR release 'emr-5.29.0', and application bundle 'Custom (Hadoop 2.8.5, Hive 2.3.6)'. It also lists the instance groups: 'Primary (m4.large), Core (m4.large), Task (m5.xlarge)'. The 'Provisioning configuration' shows 'Core size: 1 instance' and 'Task size: 1 instance'. The 'Networking - required' section is also visible.

Under the Cluster Nodes and Instances section, click the pencil icon next to the instance type for the primary node type, choose m4.large, and save.

To change the core node type instance type, click the pencil icon next to it, choose m4.large, and save.

Verify that the following numbers of instances are set:

principal node: 1

Two at the center node

Task node: 0

Leaving everything as it is, click Next.

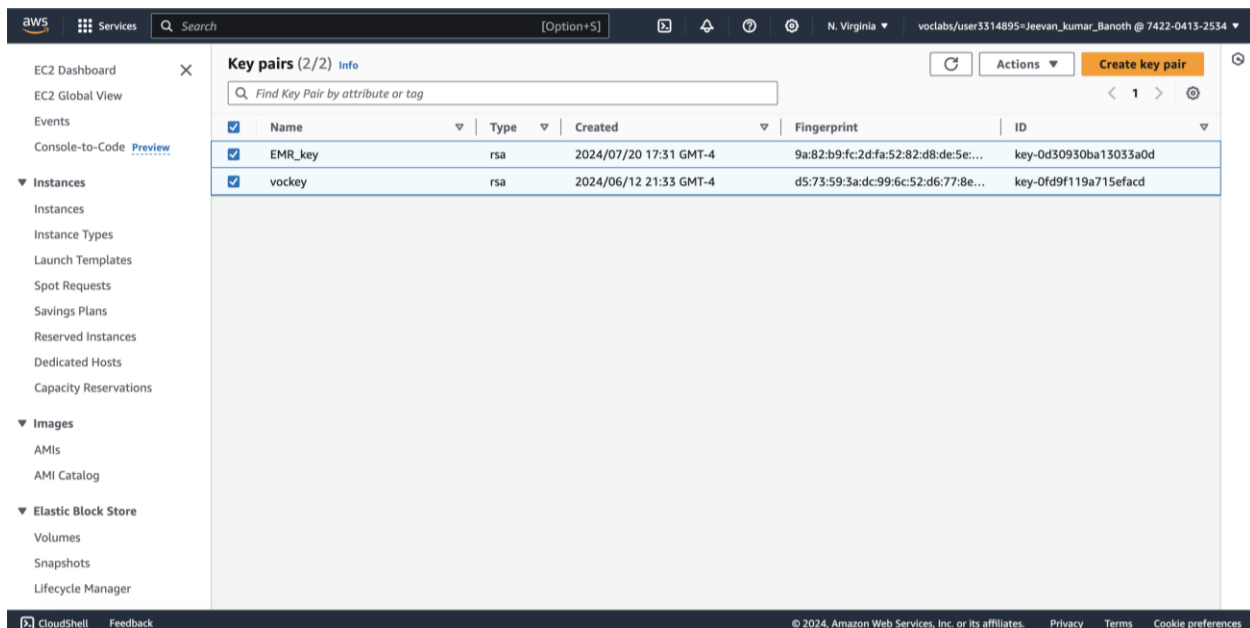
The screenshot shows the AWS EMR console interface. At the top, there's a navigation bar with the AWS logo, 'Services', a search bar, and a user profile. The main content area is titled 'Choose an option' and contains three radio buttons: 'Set cluster size manually' (selected), 'Use EMR-managed scaling', and 'Use custom automatic scaling'. Below this is the 'Provisioning configuration' section, which includes a table for instance groups. The table has columns for Name, Instance type, Instance(s) size, and Use Spot purchasing option. The 'Core' instance group is configured with 'm4.large' instance type and '1' instance size. The 'Networking - required' section is expanded, showing fields for 'Virtual private cloud (VPC)' and 'Subnet', both with 'Browse' and 'Create' buttons. A 'Steps (0)' section is also visible at the bottom.

Name	Instance type	Instance(s) size	Use Spot purchasing option
Core	m4.large	1	☐

In choose options, set cluster size manually and in the network security as shown above for VPC and subnet.

The screenshot shows the AWS EMR console interface, specifically the 'Cluster termination and node replacement' section. This section is expanded, showing 'Termination option' with three radio buttons: 'Manually terminate cluster' (selected), 'Automatically terminate cluster after last step ends', and 'Automatically terminate cluster after idle time (Recommended)'. A yellow warning box states: 'To terminate your cluster after idle time, you must choose an Amazon EMR version later than 5.30.0, except version 6.0.0.' Below this is the 'Use termination protection' section, which is currently unchecked. The 'Unhealthy node replacement' section is also expanded, showing 'Turn on' (selected) and 'Turn off' options. The 'Summary' panel on the right side of the console shows the cluster configuration details, including Name, Amazon EMR release, Application bundle, Cluster configuration, Cluster scaling and provisioning, Provisioning configuration, and Networking.

By expanding the cluster termination and node replacement section, for Termination option, select manually terminate cluster and Unhealthy node replacement turn it on.

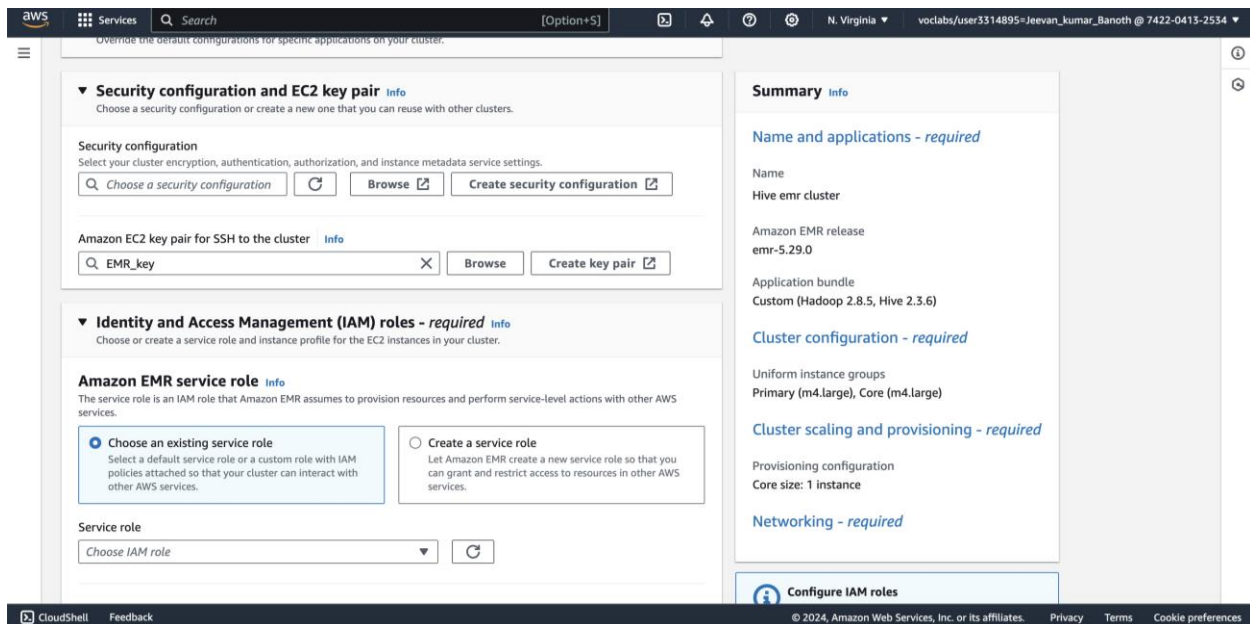


Key pairs (2/2) Info

Find Key Pair by attribute or tag

☒	Name	Type	Created	Fingerprint	ID
☒	EMR_key	rsa	2024/07/20 17:31 GMT-4	9a:82:b9:fc:2d:fa:52:82:d8:de:5e:...	key-0d30930ba13033a0d
☒	vockey	rsa	2024/06/12 21:33 GMT-4	d5:73:59:3a:dc:99:6c:52:d6:77:8e:...	key-0fd9f119a715efacd

Security Configuration: Pick the Vockey key pair for EC2 key pair. Select Custom for Permissions. Make sure that the EMR_DefaultRole is chosen for the EMR role.



Security configuration and EC2 key pair Info

Choose a security configuration or create a new one that you can reuse with other clusters.

Security configuration
Select your cluster encryption, authentication, authorization, and instance metadata service settings.

Choose a security configuration

Amazon EC2 key pair for SSH to the cluster Info

EMR_key

Identity and Access Management (IAM) roles - required Info

Choose or create a service role and instance profile for the EC2 instances in your cluster.

Amazon EMR service role Info

The service role is an IAM role that Amazon EMR assumes to provision resources and perform service-level actions with other AWS services.

☒ **Choose an existing service role**
Select a default service role or a custom role with IAM policies attached so that your cluster can interact with other AWS services.

☐ **Create a service role**
Let Amazon EMR create a new service role so that you can grant and restrict access to resources in other AWS services.

Service role
Choose IAM role

Summary Info

Name and applications - required

Name
Hive emr cluster

Amazon EMR release
emr-5.29.0

Application bundle
Custom (Hadoop 2.8.5, Hive 2.3.6)

Cluster configuration - required

Uniform instance groups
Primary (m4.large), Core (m4.large)

Cluster scaling and provisioning - required

Provisioning configuration
Core size: 1 instance

Networking - required

Make sure that the EMR_EC2_DefaultRole is chosen for the EC2 instance profile. For Auto Scaling Role, choose to continue without any role. Expand the EC2 Security groups section and use default settings.



Downloads

Today

- EMR_key.pem
- LabManual5_Proc essing...MR.docx
- DB Design 14 items

As soon as we choose the EMR_key, a .pem file will be downloaded in our local download's files as shown above.

We will use that for later in this lab.

Identity and Access Management (IAM) roles - required
Choose or create a service role and instance profile for the EC2 instances in your cluster.

Amazon EMR service role
The service role is an IAM role that Amazon EMR assumes to provision resources and perform service-level actions with other AWS services.

☒ Choose an existing service role
Select a default service role or a custom role with IAM policies attached so that your cluster can interact with other AWS services.

☐ Create a service role
Let Amazon EMR create a new service role so that you can grant and restrict access to resources in other AWS services.

Service role
LabRole

EC2 instance profile for Amazon EMR
The instance profile assigns a role to every EC2 instance in a cluster. The instance profile must specify a role that can access the resources for your steps and bootstrap actions.

☒ Choose an existing instance profile
Select a default role or a custom instance profile with IAM policies attached so that your cluster can interact with your resources in Amazon S3.

☐ Create an instance profile
Let Amazon EMR create a new instance profile so that you can specify a custom set of resources for it to access in Amazon S3.

Instance profile
EMR_EC2_DefaultRole

Summary
Name and applications - required

Name
Hive emr cluster

Amazon EMR release
emr-5.29.0

Application bundle
Custom (Hadoop 2.8.5, Hive 2.3.6)

Cluster configuration - required

Uniform instance groups
Primary (m4.large), Core (m4.large)

Cluster scaling and provisioning - required

Provisioning configuration
Core size: 1 instance

Networking - required

Cancel Create cluster

For Identity and Access management roles, we will be choosing the configuration as above.

Your cluster "Hive emr cluster" has been successfully created.

Amazon EMR > EMR on EC2: Clusters > Hive emr cluster

Hive emr cluster Updated less than a minute ago Terminate Clone in AWS CLI Clone

Summary

Cluster info	Applications	Cluster management	Status and time
Cluster ID j-26208J4TFKM19	Amazon EMR version emr-5.29.0	Log destination in Amazon S3 aws-logs-742204132534-us-east-1/elasticmapreduce	Status Starting
Cluster configuration Instance groups	Installed applications Hadoop 2.8.5, Hive 2.3.6	Primary node public DNS -	Creation time July 20, 2024, 19:16 (UTC-04:00)
Capacity 1 Primary 1 Core 0 Task			Elapsed time 0 seconds

Properties Bootstrap actions Instances (Hardware) Steps Applications Configurations Monitoring Events Tags (0)

Cluster logs Info

Archive log files to Amazon S3
Turned on

Amazon S3 location
s3://aws-logs-742204132534-us-east-

Encryption for logs
Turned off

Cluster termination and node replacement Info Edit

Termination option
Manually terminate cluster

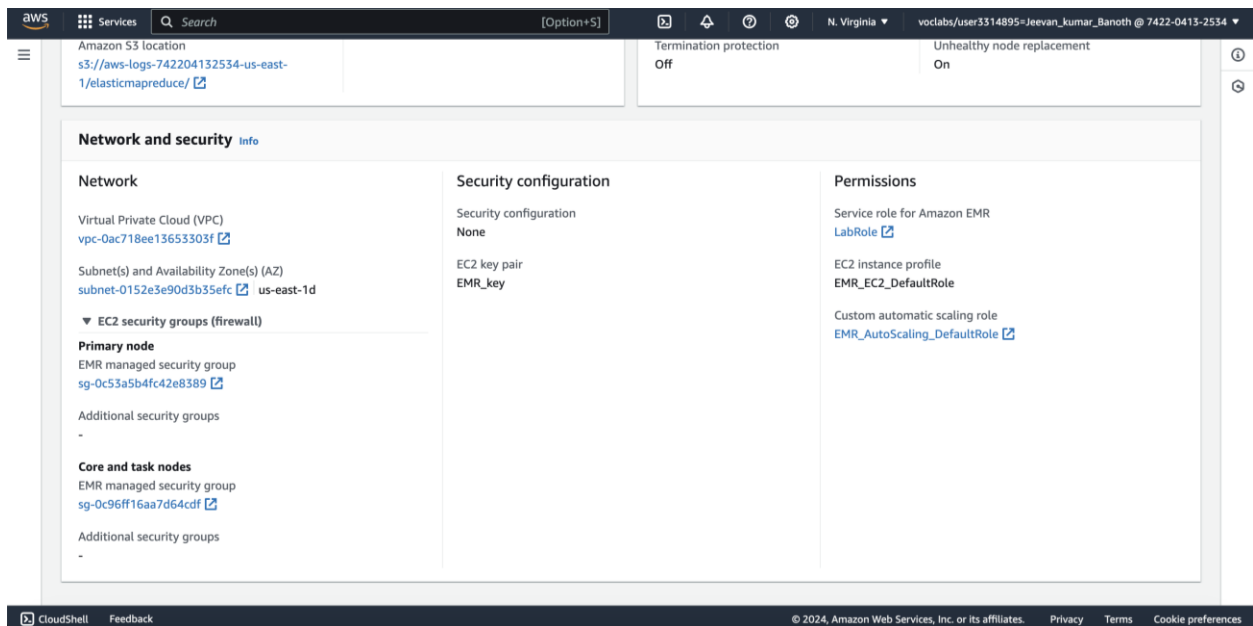
Termination protection
Off

Idle time
-

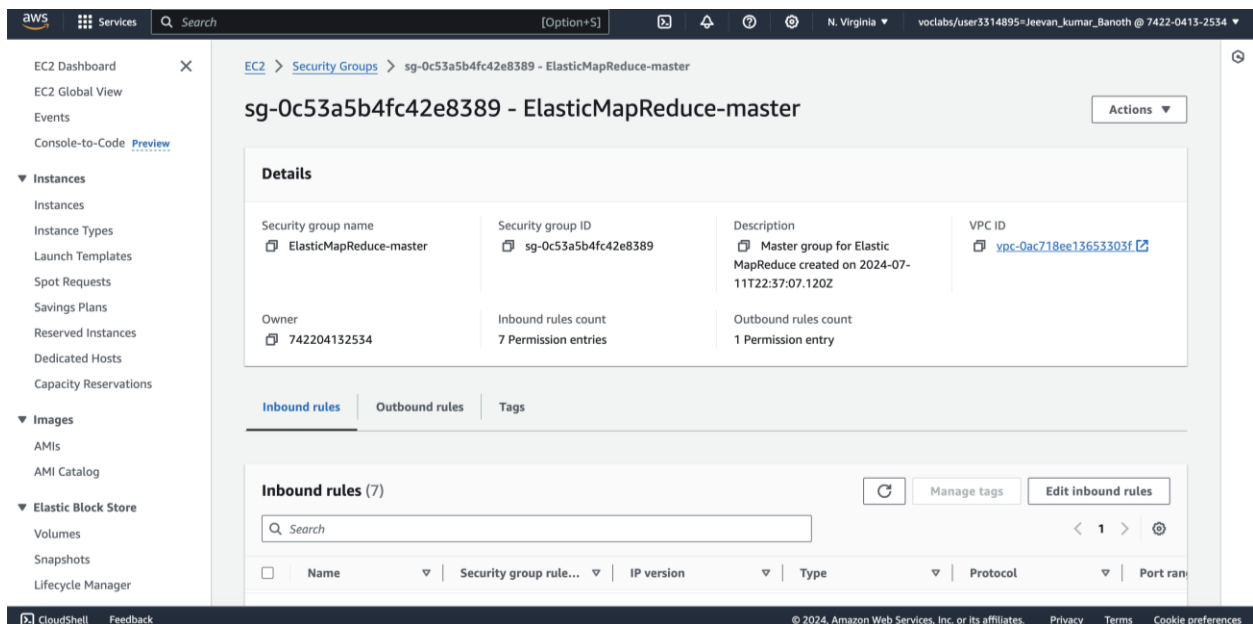
Unhealthy node replacement
On

Then press create cluster, it will be in the starting status as shown above.

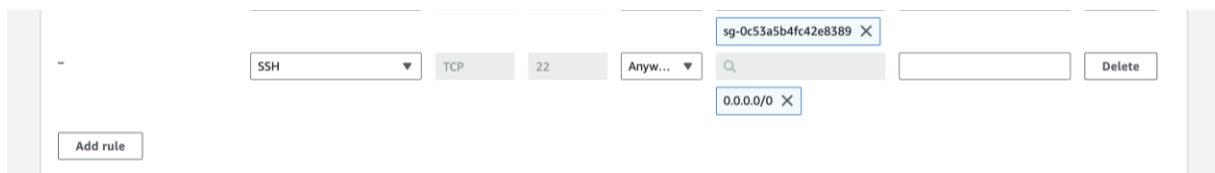
In the Network and security tab, we can see a primary node and there is a link under it, usually it starts with sg-0 something like this, as shown below.



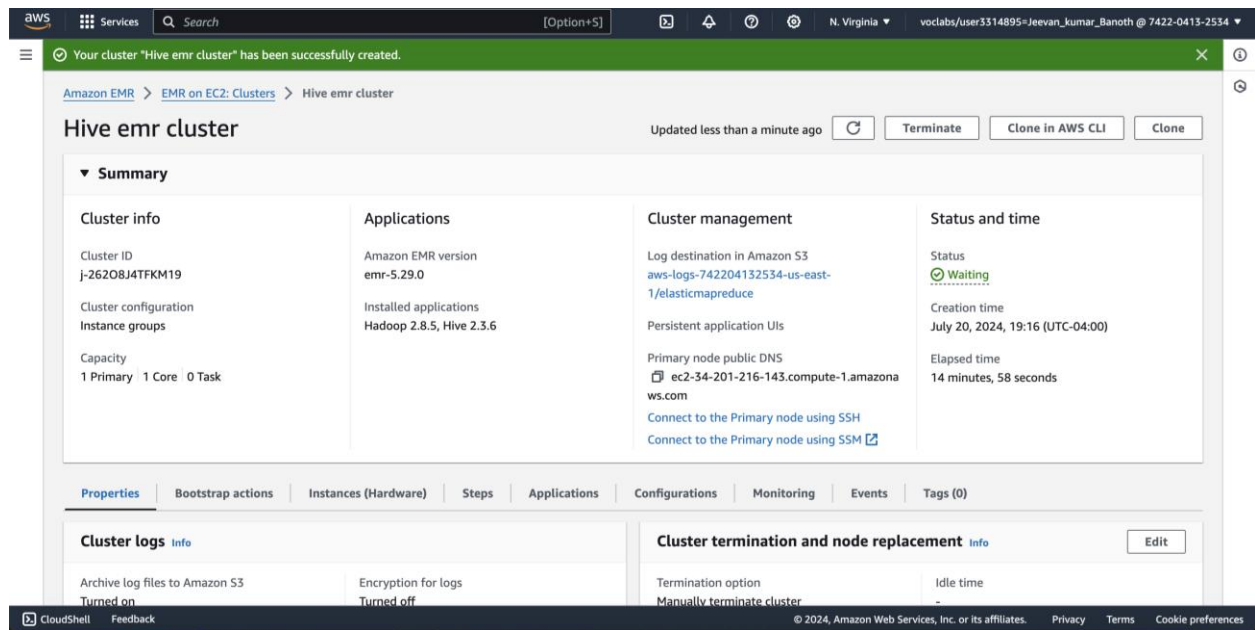
Click on that link, it will open in a different tab, as shown below.



Here, choose in the inbound rules as edit inbound rules and add rule in there as below.



In the place of type, choose SSH and followed by Anywhere IPv4 and save it. This gets added to our security group.

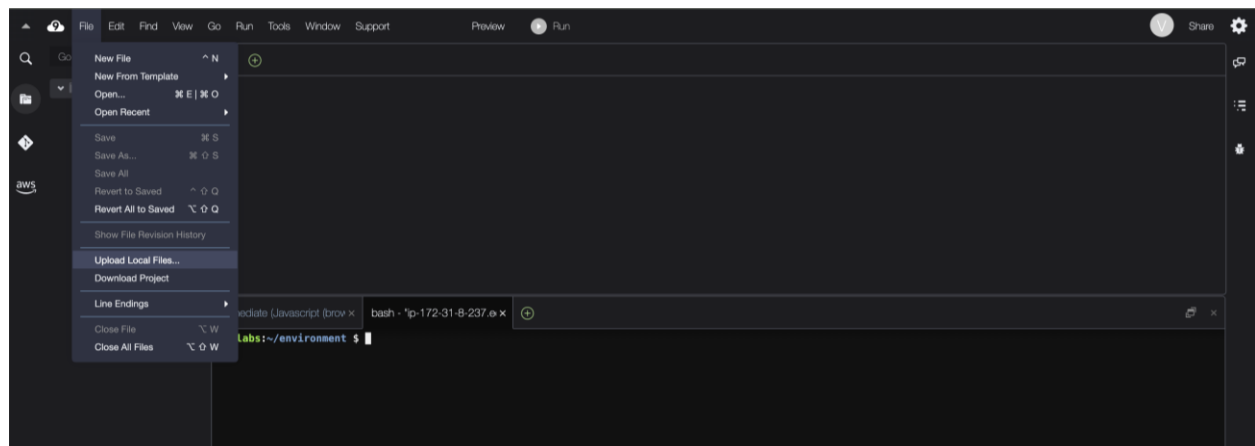


Now, as you can see above status get changes to waiting.

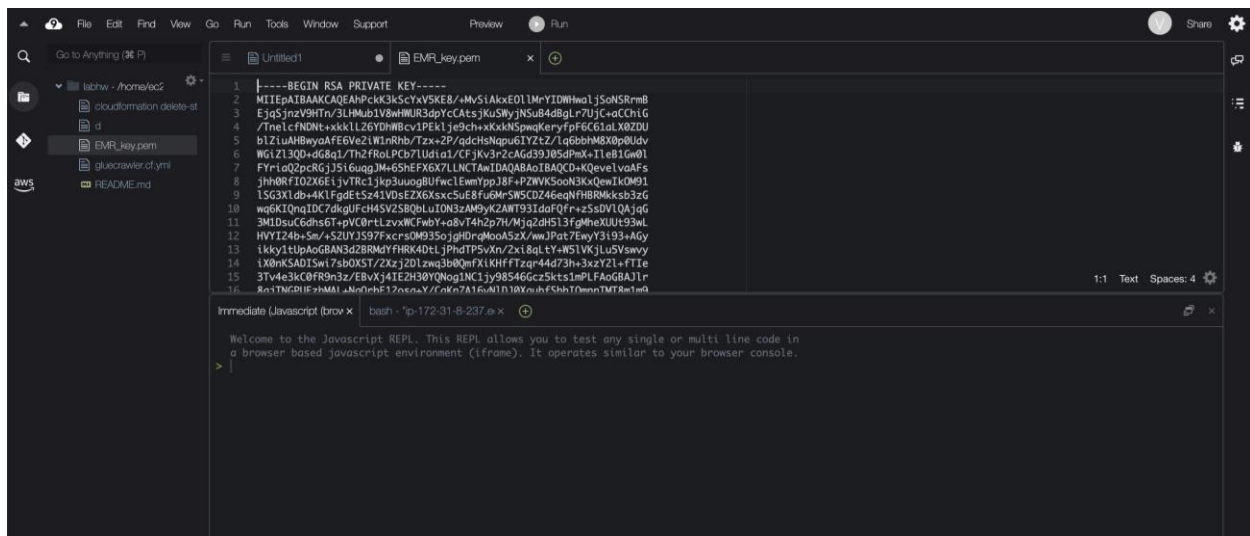
Task 2: Remote Connection to Hadoop Master via SSH

Locate the public DNS address assigned to the main node in the Cluster Summary tab and make a copy.

Next, go to your AWS Management Console and search for Cloud9 to open it. You will now select the Cloud9 Instance environment, then click on Open IDE. Now you create a new file from Cloud9 IDE by choosing File > New File. Lastly, paste this public DNS address in a document so you can use it later.

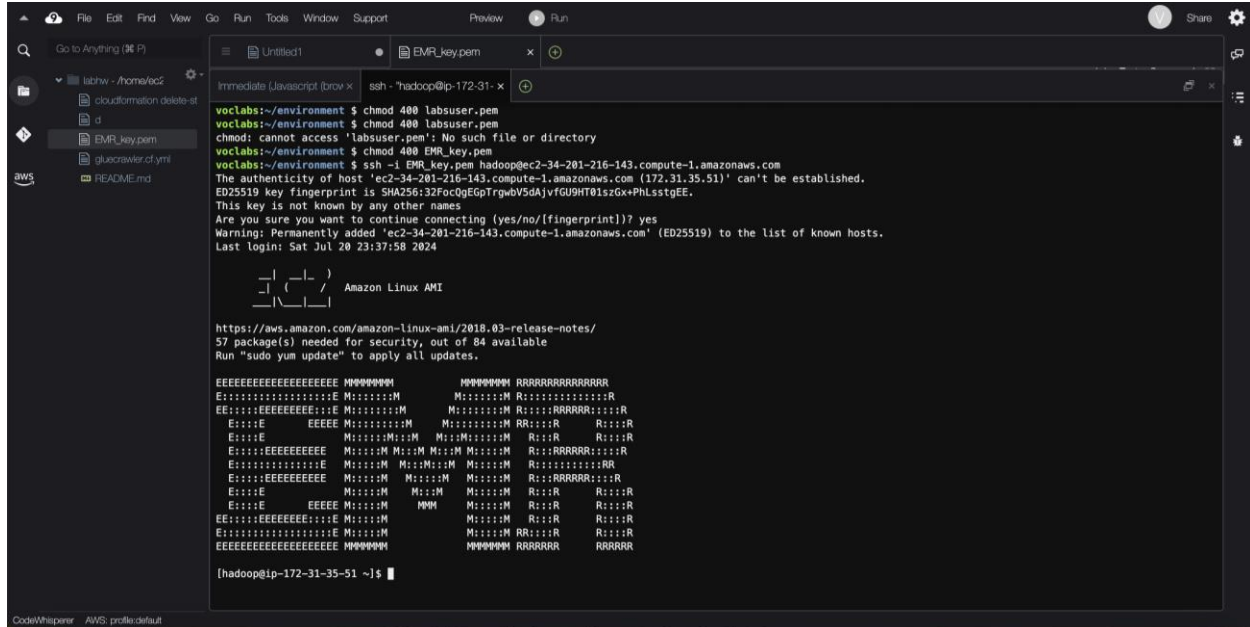


As you can see above, there is a upload button, by using that we will going to upload the EMR_Key.pem file which we have downloaded earlier.



```
1 -----BEGIN RSA PRIVATE KEY-----
2 MIIEpATBAKCAQEAPcK3K5CYV5KE8/+MvS1AkxE01IMrYIDWw01jSoNSRrmB
3 Eja5jnzV9HTr/3LHMub1V8wHUR3dpYcCAtsJKuSMyjNSuB4d8gLn7UJC+aCChIG
4 /TnelcFN0nt+xxkL126YDHWBcv1PEK1je9ch+XKxKNSpwKeryfpF6CG1aLX0ZDU
5 bLZ1uAHBmyaAF6Ve2LWlnRhb/Tzx+2P/qdchSnpu6LYZtZ/Lq6bthM8X0p0Udv
6 W61Z1300+qG8a1/Tk2/Ro1PC871Uda1a1CFJk3p2zGd9308p0K+11e1Gw01
7 FYr1aQ2pC8Gj3516uagJW+6SHEFXG7LUNCTAwIDAQA8AotBAQCD+KQevelvaAFs
8 jHHORFIO2X6E1jvTRc1jKp3uagBUfwclEwmYp18F+PZWVKSooN3KxQewIk0M91
9 1SG3X1db+4K1FgdEtSz41VDsEZ6Xsxc5uE8Fu6M+SW5CD246eqNFHBRMkksb3zG
10 wq6K1Qnq1DC7dkgUfch45V2S8QbLu10N3zAMByK2ANT93IdaFQfr+z5dV1QAjg6
11 3M1Dsuc6dhs6T+pV08tLzvXWCFwbY+q8vT4h2p7H/Mja2dH513f9mheXUUt93wL
12 HWY124b+Sm/+SZUY1S97FxcrsDM035ojghDngMoa5Zx/wwJPat7eyY3193+AGy
13 1kky1t1lpAGGdAN3d2BRM4YFHRK4d11jnuDPSuXru2x184tY+NS1KJ1uVowvy
14 1X0nKSAD1Sw17abOKST/2xzj2D1zw3b0QmFxiKHffTzqr44d73h+3xzy21+FTIE
15 3Tv4e3kC0FR9n3z/EBvXj4IE2H30YONog1NC1jy98546Gcz5kts1mPLFa0GBAJ1r
16 Rn1TNGRIE+HMAI+No0ehf12nsaY/cnKn7A16uK110YnshFShhT0mnTMT8m1m0
```

As you can see the file has been uploaded to cloud9 successfully.



```
Immediate Javascript (brow x) ssh -i hadoop@ip-172-31- x
Welcome to the Javascript REPL. This REPL allows you to test any single or multi line code in
a browser based javascript environment (iframe). It operates similar to your browser console.
>

voclabs:~/environment $ chmod 400 labsuser.pem
voclabs:~/environment $ chown labsuser:labsuser labsuser.pem
chmod: cannot access 'labsuser.pem': No such file or directory
voclabs:~/environment $ chmod 400 EMR_key.pem
voclabs:~/environment $ ssh -i EMR_key.pem hadoop@ec2-34-201-216-143.compute-1.amazonaws.com
The authenticity of host 'ec2-34-201-216-143.compute-1.amazonaws.com (172.31.35.51)' can't be established.
ED25519 key fingerprint is SHA256:32FocQeGpTrgwBV5dAjvFGU9HT0zszGx+PHLstgEE.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-34-201-216-143.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
Last login: Sat Jul 20 23:37:58 2024

 _ _ _ _ _
| | | | |
|_|_|_|_|_|

Amazon Linux AMI

https://aws.amazon.com/amazon-linux-ami/2018.03-release-notes/
57 package(s) needed for security, out of 84 available
Run "sudo yum update" to apply all updates.

EEEEEEEEEEEEEEEEEEEE MMMMMMM RRRRRRRRRRRRRRR
Ei:::iiiiiiiiiiiiiiE M:::iiii M:::iiii R:::iiiiiiiiiiiiR
EE:::iiEEEEEEEEEEEE E M:::iiii M:::iiii R:::iiRRRRRR:::iiR
E:::iiE EEEEE M:::iiiiii M:::iiiiii RR:::iiR R:::iiR
E:::iiE M:::iiii M:::ii M:::iiii M:::ii R:::iiR R:::iiR
E:::iiEEEEEEEEEE M:::ii M:::ii M:::ii M:::ii R:::iiRRRRR:::iiR
E:::iiEEEEEEEEEE M:::ii M:::ii M:::ii M:::ii R:::iiiiiiRR
E:::iiEEEEEEEEEE M:::ii M:::ii M:::ii R:::iiRRRRR:::iiR
E:::iiE M:::ii M:::ii M:::ii M:::ii R:::iiR R:::iiR
E:::iiE EEEEE M:::ii M:::ii M:::ii M:::ii R:::iiR R:::iiR
EE:::iiEEEEEEEEEE E M:::ii M:::ii M:::ii M:::ii R:::iiR
E:::iiiiEEEEEEEEEE E M:::ii M:::ii M:::ii M:::ii R:::iiR
EEEEEEEEEEEEEEEEEEEE MMMMMMM RRRRRRR RRRRRR

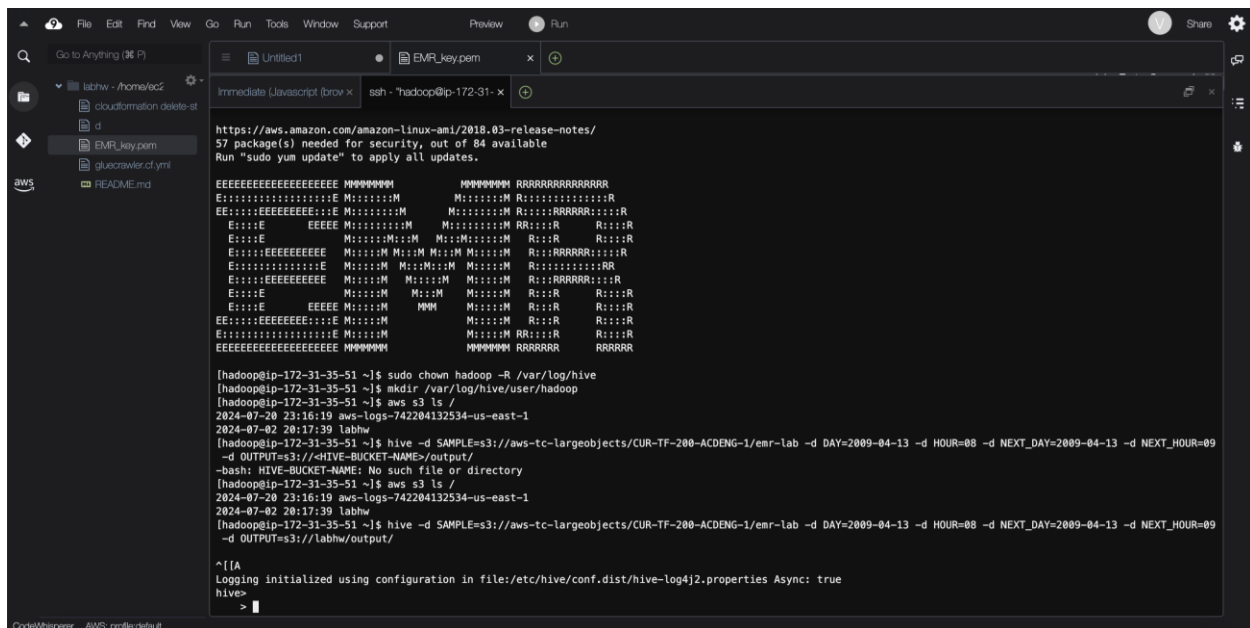
[hadoop@ip-172-31-35-51 ~]$
```

Now we will start examining the file by using the commands given in the lab manual.

First, we will be changing the directory to EMR_key.pem file because that's the file we are examining so.

Next, we will be computing it by using the ssh commands given by replacing the file name with what we gave.

In my case it is EMR_key.pem so We must change this according to the code.



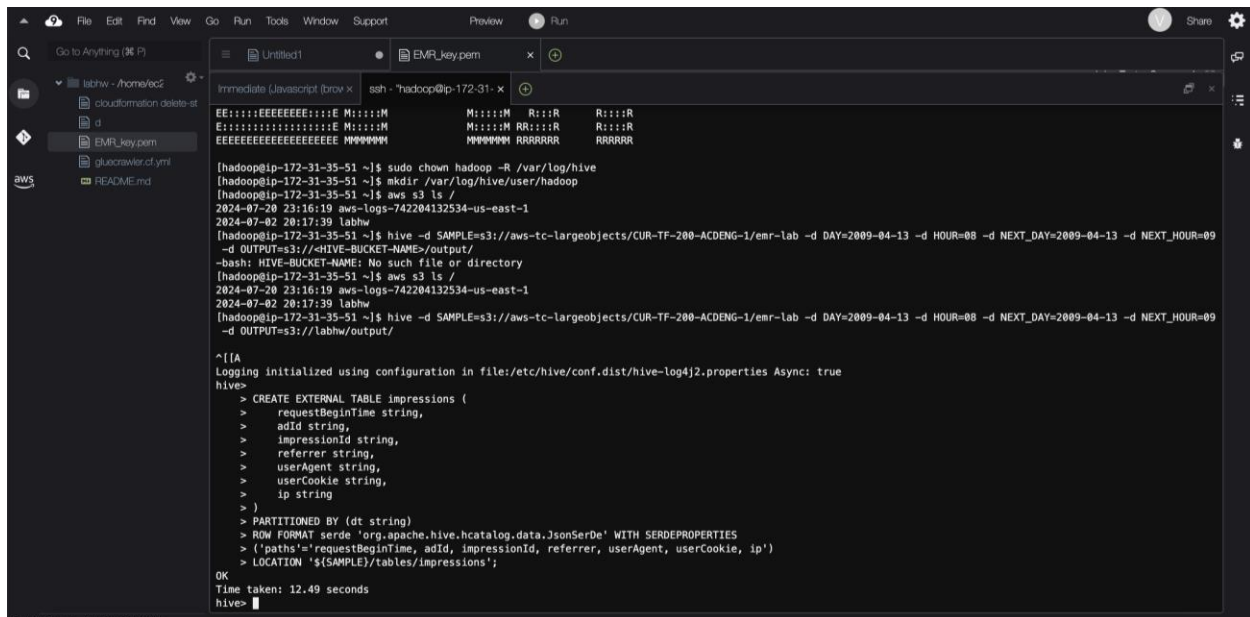
```
https://aws.amazon.com/amazon-linux-ami/2018.03-release-notes/
57 package(s) needed for security, out of 84 available
Run "sudo yum update" to apply all updates.

EEEEEEEEEEEEEEEEEEEE MMMMMMM RRRRRRRRRRRRRRR
E::::::::::::::::::::E M::::::::M M::::::::M R:::::::::R
EE::::::::::::::::::::E M::::::::M M::::::::M R:::::::::R
E::::E EEEEE M::::::::M M::::::::M RR::::R R::::R
E::::E M::::::::M M::::::::M R::::R R::::R
E::::::::::::::::::::E M::::::::M M::::::::M R:::::::::R
E::::::::::::::::::::E M::::::::M M::::::::M R:::::::::R
E::::::::::::::::::::E M::::::::M M::::::::M R:::::::::R
E::::E M::::::::M M::::::::M R::::R R::::R
E::::E EEEEE M::::::::M M M::::::::M R::::R R::::R
EE::::::::::::::::::::E M::::::::M M::::::::M R::::R R::::R
E::::::::::::::::::::E M::::::::M M::::::::M RR::::R R::::R
EEEEEEEEEEEEEEEEEEEE MMMMMMM RRRRRRRRRRRRRRR

[hadoopip-172-31-35-51 ~]$ sudo chown hadoop -R /var/log/hive
[hadoopip-172-31-35-51 ~]$ mkdir /var/log/hive/user/hadoop
[hadoopip-172-31-35-51 ~]$ aws s3 ls /
2024-07-20 23:16:19 aws-logs-742204132534-us-east-1
2024-07-20 20:17:39 labhw
[hadoopip-172-31-35-51 ~]$ hive -d SAMPLE=s3://aws-logs-742204132534-us-east-1
-d OUTPUT=s3://HIVE-BUCKET-NAME/output/
-bash: HIVE-BUCKET-NAME: No such file or directory
[hadoopip-172-31-35-51 ~]$ aws s3 ls /
2024-07-20 23:16:19 aws-logs-742204132534-us-east-1
2024-07-20 20:17:39 labhw
[hadoopip-172-31-35-51 ~]$ hive -d SAMPLE=s3://aws-logs-742204132534-us-east-1
-d OUTPUT=s3://labhw/output/

^[[A
Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j2.properties Async: true
hive>
>
```

By using aws and hive commands we will find the bucket name that we are working on and replace that with the next code.



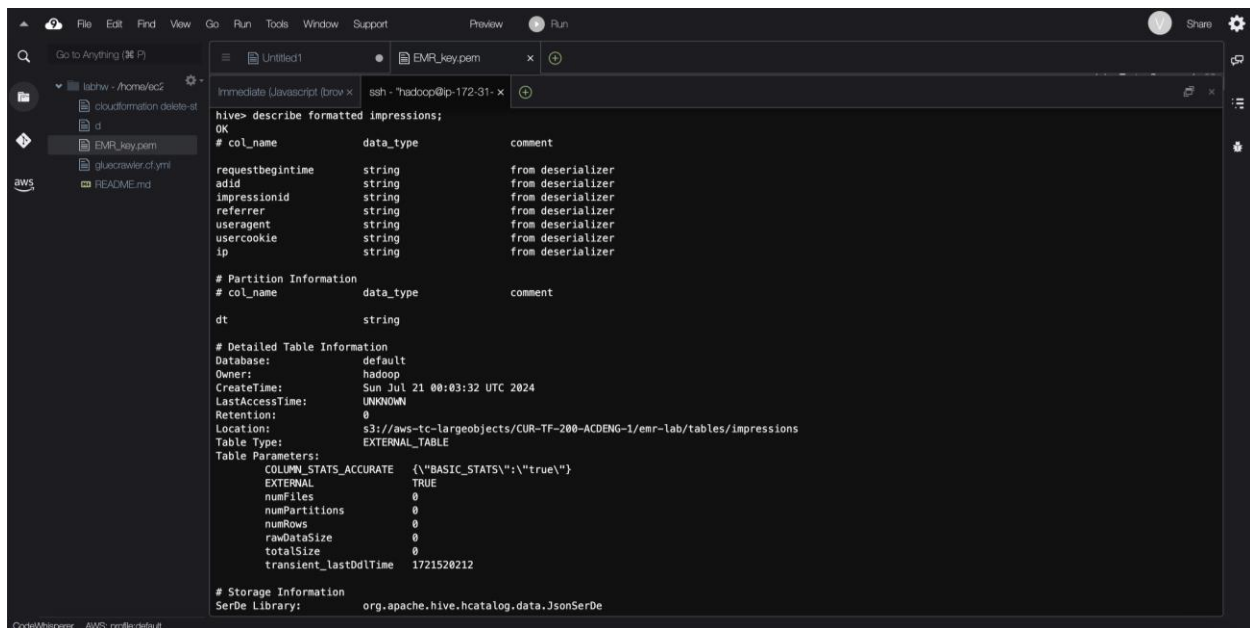
```
[hadoopip-172-31-35-51 ~]$ sudo chown hadoop -R /var/log/hive
[hadoopip-172-31-35-51 ~]$ mkdir /var/log/hive/user/hadoop
[hadoopip-172-31-35-51 ~]$ aws s3 ls /
2024-07-20 23:16:19 aws-logs-742204132534-us-east-1
2024-07-20 20:17:39 labhw
[hadoopip-172-31-35-51 ~]$ hive -d SAMPLE=s3://aws-logs-742204132534-us-east-1
-d OUTPUT=s3://HIVE-BUCKET-NAME/output/
-bash: HIVE-BUCKET-NAME: No such file or directory
[hadoopip-172-31-35-51 ~]$ aws s3 ls /
2024-07-20 23:16:19 aws-logs-742204132534-us-east-1
2024-07-20 20:17:39 labhw
[hadoopip-172-31-35-51 ~]$ hive -d SAMPLE=s3://aws-logs-742204132534-us-east-1
-d OUTPUT=s3://labhw/output/

^[[A
Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j2.properties Async: true
hive>
> CREATE EXTERNAL TABLE impressions (
>   requestBeginTime string,
>   adId string,
>   impressionId string,
>   referrer string,
>   userAgent string,
>   userCookie string,
>   ip string
> )
> PARTITIONED BY (dt string)
> ROW FORMAT serde 'org.apache.hive.hcatalog.data.JsonSerDe' WITH SERDEPROPERTIES
> ('paths'='requestBeginTime, adId, impressionId, referrer, userAgent, userCookie, ip')
> LOCATION '${SAMPLE}/tables/impressions';

OK
Time taken: 12.49 seconds
hive>
```

So, from that hive -d command, we will be going to be redirected into the hive terminal and we will work in that only from now on.

Next code is for creating the impressions table and time taken to run this is 12.49 seconds.



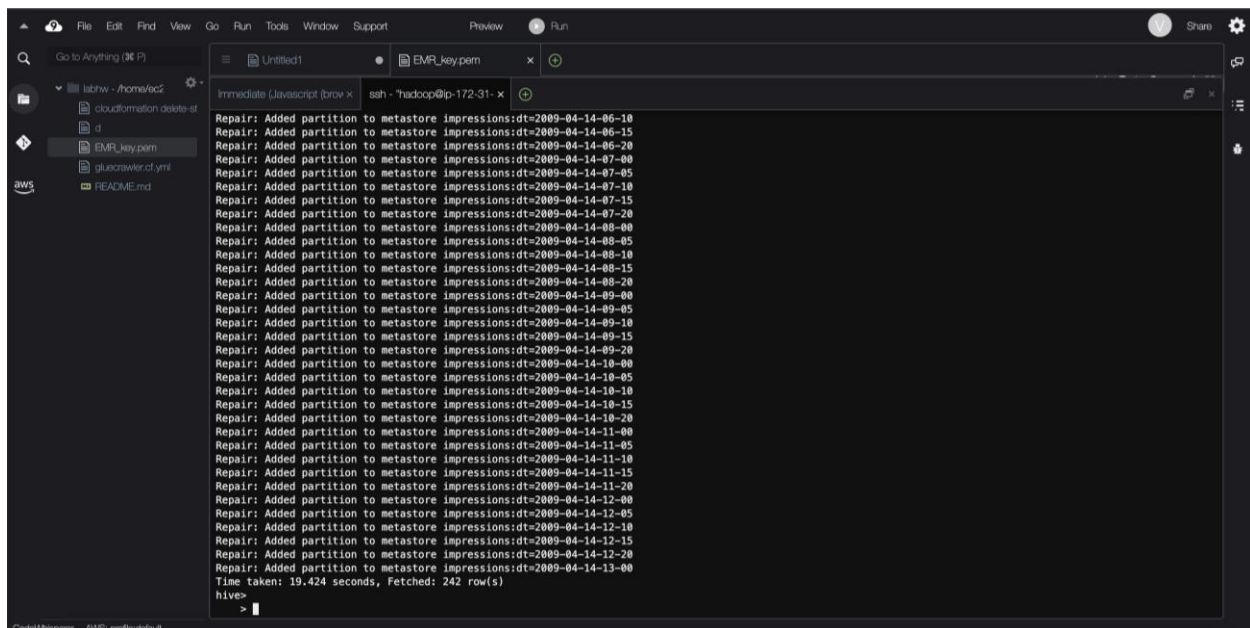
```
hive> describe formatted impressions;
OK
# col_name          data_type          comment
requestbeginTime    string              from deserializer
addid                string              from deserializer
impressionid         string              from deserializer
referrer             string              from deserializer
useragent            string              from deserializer
usercookie           string              from deserializer
ip                   string              from deserializer

# Partition Information
# col_name          data_type          comment
dt                   string

# Detailed Table Information
Database:            default
Owner:               hadoop
CreateTime:          Sun Jul 21 00:03:32 UTC 2024
LastAccessTime:      UNKNOWN
Retention:           0
Location:             s3://aws-tc-largeobjects/CUR-TF-200-ACDENG-1/emr-lab/tables/impressions
Table Type:          EXTERNAL_TABLE
Table Parameters:
COLUMN_STATS_ACCURATE {\"BASIC_STATS\":true}
EXTERNAL              TRUE
numFiles              0
numPartitions         0
numRows               0
rawDataSize           0
totalSize              0
transient_lastDdlTime 1721520212

# Storage Information
SerDe Library:        org.apache.hive.hcatalog.data.JsonSerDe
```

Next we will be updating the hive metadata for impressions by using the commands. Like for checking current partitions, repair table to add partitions and verifying the partitions.



```
hive> repair table impressions;
Repair: Added partition to metastore impressions:dt=2009-04-14-06-10
Repair: Added partition to metastore impressions:dt=2009-04-14-06-15
Repair: Added partition to metastore impressions:dt=2009-04-14-06-20
Repair: Added partition to metastore impressions:dt=2009-04-14-07-00
Repair: Added partition to metastore impressions:dt=2009-04-14-07-05
Repair: Added partition to metastore impressions:dt=2009-04-14-07-10
Repair: Added partition to metastore impressions:dt=2009-04-14-07-15
Repair: Added partition to metastore impressions:dt=2009-04-14-07-20
Repair: Added partition to metastore impressions:dt=2009-04-14-08-00
Repair: Added partition to metastore impressions:dt=2009-04-14-08-05
Repair: Added partition to metastore impressions:dt=2009-04-14-08-10
Repair: Added partition to metastore impressions:dt=2009-04-14-08-15
Repair: Added partition to metastore impressions:dt=2009-04-14-08-20
Repair: Added partition to metastore impressions:dt=2009-04-14-09-00
Repair: Added partition to metastore impressions:dt=2009-04-14-09-05
Repair: Added partition to metastore impressions:dt=2009-04-14-09-10
Repair: Added partition to metastore impressions:dt=2009-04-14-09-15
Repair: Added partition to metastore impressions:dt=2009-04-14-09-20
Repair: Added partition to metastore impressions:dt=2009-04-14-10-00
Repair: Added partition to metastore impressions:dt=2009-04-14-10-05
Repair: Added partition to metastore impressions:dt=2009-04-14-10-10
Repair: Added partition to metastore impressions:dt=2009-04-14-10-15
Repair: Added partition to metastore impressions:dt=2009-04-14-10-20
Repair: Added partition to metastore impressions:dt=2009-04-14-11-00
Repair: Added partition to metastore impressions:dt=2009-04-14-11-05
Repair: Added partition to metastore impressions:dt=2009-04-14-11-10
Repair: Added partition to metastore impressions:dt=2009-04-14-11-15
Repair: Added partition to metastore impressions:dt=2009-04-14-11-20
Repair: Added partition to metastore impressions:dt=2009-04-14-12-00
Repair: Added partition to metastore impressions:dt=2009-04-14-12-05
Repair: Added partition to metastore impressions:dt=2009-04-14-12-10
Repair: Added partition to metastore impressions:dt=2009-04-14-12-15
Repair: Added partition to metastore impressions:dt=2009-04-14-12-20
Repair: Added partition to metastore impressions:dt=2009-04-14-13-00
Time taken: 19.424 seconds, Fetched: 242 row(s)
hive>
```

Next, we will be creating an external table for clicks.

```
File Edit Find View Go Run Tools Window Support Preview Run Share
Go to Anything (⌘ P)
labnw - /home/lab2
cloudformation delete-st
d
EMR_key.pem
gluecrawler.d.yml
aws
README.md

Untitled1
EMR_key.pem
ssh - "hadoop@ip-172-31-1-1"
Immediate (Javascript (brow x)
ip
string
from deserializer
# Partition Information
# col_name
data_type
comment
dt
string
# Detailed Table Information
Database: default
Owner: hadoop
CreateTime: Sun Jul 21 00:03:32 UTC 2024
LastAccessTime: UNKNOWN
Retention: 0
Location: s3://aws-tc-largeobjects/CUR-TF-200-ACDENG-1/emr-lab/tables/impressions
Table Type: EXTERNAL_TABLE
Table Parameters:
EXTERNAL TRUE
numFiles 1446
numPartitions 241
numRows 0
rawDataSize 0
totalSize 650017289
transient_lastDdlTime 1721520212
# Storage Information
SerDe Library: org.apache.hive.hcatalog.data.JsonSerDe
InputFormat: org.apache.hadoop.mapred.TextInputFormat
OutputFormat: org.apache.hadoop.hive ql.io.HiveIgnoreKeyTextOutputFormat
Compressed: No
Num Buckets: -1
Bucket Columns: []
Sort Columns: []
Storage Desc Params:
paths requestBeginTime, adId, impressionId, referrer, userAgent, userCookie, ip
serialization.format 1
Time taken: 0.244 seconds, Fetched: 43 row(s)
hive>
```

Now, in this we will be updating the hive metadata for clicks and repairing the table to add partitions.

```
File Edit Find View Go Run Tools Window Support Preview Run Share
Go to Anything (⌘ P)
labnw - /home/lab2
cloudformation delete-st
d
EMR_key.pem
gluecrawler.d.yml
aws
README.md

Untitled1
EMR_key.pem
ssh - "hadoop@ip-172-31-1-1"
Immediate (Javascript (brow x)
hive> CREATE EXTERNAL TABLE clicks (impressionId string, adId string)
> PARTITIONED BY (dt string)
> ROW FORMAT serde 'org.apache.hive.hcatalog.data.JsonSerDe'
> WITH SERDEPROPERTIES ('paths'='impressionId')
> LOCATION '${SAMPLE}/tables/clicks';
OK
Time taken: 0.57 seconds
hive> MSCK REPAIR TABLE clicks;
OK
Partitions not in metastore: clicks:dt=2009-04-12-13-00 clicks:dt=2009-04-12-13-05 clicks:dt=2009-04-12-13-10 clicks:dt=2009-04-12-13-15 cli
cks:dt=2009-04-12-13-20 clicks:dt=2009-04-12-14-00 clicks:dt=2009-04-12-14-05 clicks:dt=2009-04-12-14-10 clicks:dt=2009-04-12-14-15 clicks:dt=2
009-04-12-14-20 clicks:dt=2009-04-12-15-00 clicks:dt=2009-04-12-15-05 clicks:dt=2009-04-12-15-10 clicks:dt=2009-04-12-15-15 clicks:dt=2009-04-1
2-15-20 clicks:dt=2009-04-12-16-00 clicks:dt=2009-04-12-16-05 clicks:dt=2009-04-12-16-10 clicks:dt=2009-04-12-16-15 clicks:dt=2009-04-12-16-20c
licks:dt=2009-04-12-17-00 clicks:dt=2009-04-12-17-05 clicks:dt=2009-04-12-17-10 clicks:dt=2009-04-12-17-15 clicks:dt=2009-04-12-17-20 cli
cks:dt=2009-04-12-18-00 clicks:dt=2009-04-12-18-05 clicks:dt=2009-04-12-18-10 clicks:dt=2009-04-12-18-15 clicks:dt=2009-04-12-18-20 clicks:dt=2
009-04-12-19-00 clicks:dt=2009-04-12-19-05 clicks:dt=2009-04-12-19-10 clicks:dt=2009-04-12-19-15 clicks:dt=2009-04-12-19-20 clicks:dt=2009-04-1
2-20-00 clicks:dt=2009-04-12-20-05 clicks:dt=2009-04-12-20-10 clicks:dt=2009-04-12-20-15 clicks:dt=2009-04-12-20-20 clicks:dt=2009-04-12-21-00c
licks:dt=2009-04-12-21-05 clicks:dt=2009-04-12-21-10 clicks:dt=2009-04-12-21-15 clicks:dt=2009-04-12-21-20 clicks:dt=2009-04-12-22-00 cli
cks:dt=2009-04-12-22-05 clicks:dt=2009-04-12-22-10 clicks:dt=2009-04-12-22-15 clicks:dt=2009-04-12-22-20 clicks:dt=2009-04-12-23-00 clicks:dt=2
009-04-12-23-05 clicks:dt=2009-04-12-23-10 clicks:dt=2009-04-12-23-15 clicks:dt=2009-04-12-23-20 clicks:dt=2009-04-13-00-00 clicks:dt=2009-04-1
3-00-05 clicks:dt=2009-04-13-01-00 clicks:dt=2009-04-13-01-05 clicks:dt=2009-04-13-01-10 clicks:dt=2009-04-13-01-15 clicks:dt=2009-04-13-01-20c
licks:dt=2009-04-13-01-25 clicks:dt=2009-04-13-02-00 clicks:dt=2009-04-13-02-05 clicks:dt=2009-04-13-02-10 clicks:dt=2009-04-13-02-15 clicks:dt=2
009-04-13-03-00 clicks:dt=2009-04-13-03-05 clicks:dt=2009-04-13-03-10 clicks:dt=2009-04-13-03-15 clicks:dt=2009-04-13-03-20 clicks:dt=2009-04-13-04-00
clicks:dt=2009-04-13-04-05 clicks:dt=2009-04-13-04-10 clicks:dt=2009-04-13-04-15 clicks:dt=2009-04-13-04-20 clicks:dt=2009-04-13-05-00 clicks:dt=2009-04-13-05-05
clicks:dt=2009-04-13-05-10 clicks:dt=2009-04-13-05-15 clicks:dt=2009-04-13-05-20 clicks:dt=2009-04-13-06-00 clicks:dt=2009-04-13-06-05 clicks:dt=2009-04-13-06-10
clicks:dt=2009-04-13-06-15 clicks:dt=2009-04-13-06-20 clicks:dt=2009-04-13-07-00 clicks:dt=2009-04-13-07-05 clicks:dt=2009-04-13-07-10 clicks:dt=2
009-04-13-07-15 clicks:dt=2009-04-13-07-20 clicks:dt=2009-04-13-08-00 clicks:dt=2009-04-13-08-05 clicks:dt=2009-04-13-08-10 clicks:dt=2009-04-13-08-15
clicks:dt=2009-04-13-09-00 clicks:dt=2009-04-13-09-05 clicks:dt=2009-04-13-09-10 clicks:dt=2009-04-13-09-15 clicks:dt=2009-04-13-09-20 clicks:dt=2009-04-13-09-25
clicks:dt=2009-04-13-10-00 clicks:dt=2009-04-13-10-05 clicks:dt=2009-04-13-10-10 clicks:dt=2009-04-13-10-15 clicks:dt=2009-04-13-11-00 clicks:dt=2009-04-13-11-05
clicks:dt=2009-04-13-11-10 clicks:dt=2009-04-13-11-15 clicks:dt=2009-04-13-12-00 clicks:dt=2009-04-13-12-05 clicks:dt=2009-04-13-12-10 clicks:dt=2009-04-13-12-15
clicks:dt=2009-04-13-12-20 clicks:dt=2009-04-13-13-00 clicks:dt=2009-04-13-13-05 clicks:dt=2009-04-13-13-10 clicks:dt=2009-04-13-13-15 clicks:dt=2009-04-13-13-20c
licks:dt=2009-04-13-14-00 clicks:dt=2009-04-13-14-05 clicks:dt=2009-04-13-14-10 clicks:dt=2009-04-13-14-15 clicks:dt=2009-04-13-14-20 cli
cks:dt=2009-04-13-15-00 clicks:dt=2009-04-13-15-05 clicks:dt=2009-04-13-15-10 clicks:dt=2009-04-13-15-15 clicks:dt=2009-04-13-15-20 clicks:dt=2
009-04-13-16-00 clicks:dt=2009-04-13-16-05 clicks:dt=2009-04-13-16-10 clicks:dt=2009-04-13-16-15 clicks:dt=2009-04-13-16-20 clicks:dt=2009-04-1
3-17-00 clicks:dt=2009-04-13-17-05 clicks:dt=2009-04-13-17-10 clicks:dt=2009-04-13-17-15 clicks:dt=2009-04-13-17-20 clicks:dt=2009-04-13-18-00c
```

Then we will have to verify the tables. Like listing all the tables and describing the click in table.

```
hive> show tables;
OK
clicks
impressions
Time taken: 0.092 seconds, Fetched: 2 row(s)
hive> describe formatted clicks;
OK
# col_name           data_type           comment
impressionid         string              from deserializer
adid                  string              from deserializer
# Partition Information
# col_name           data_type           comment
dt                    string
# Detailed Table Information
Database:              default
Owner:                 hadoop
CreateTime:            Sun Jul 21 00:05:15 UTC 2024
LastAccessTime:        UNKNOWN
Retention:              0
Location:              s3://aws-tc-largeobjects/CUR-TF-200-ACDENG-1/emr-lab/tables/clicks
Table Type:            EXTERNAL_TABLE
Table Parameters:
EXTERNAL              TRUE
numFiles              1115
numPartitions         186
numRows               0
rawDataSize           0
totalSize             23864599
transient_lastDdlTime 1721520315
# Storage Information
SerDe Library:         org.apache.hive.hcatalog.data.JsonSerDe
InputFormat:           org.apache.hadoop.mapred.TextInputFormat
```

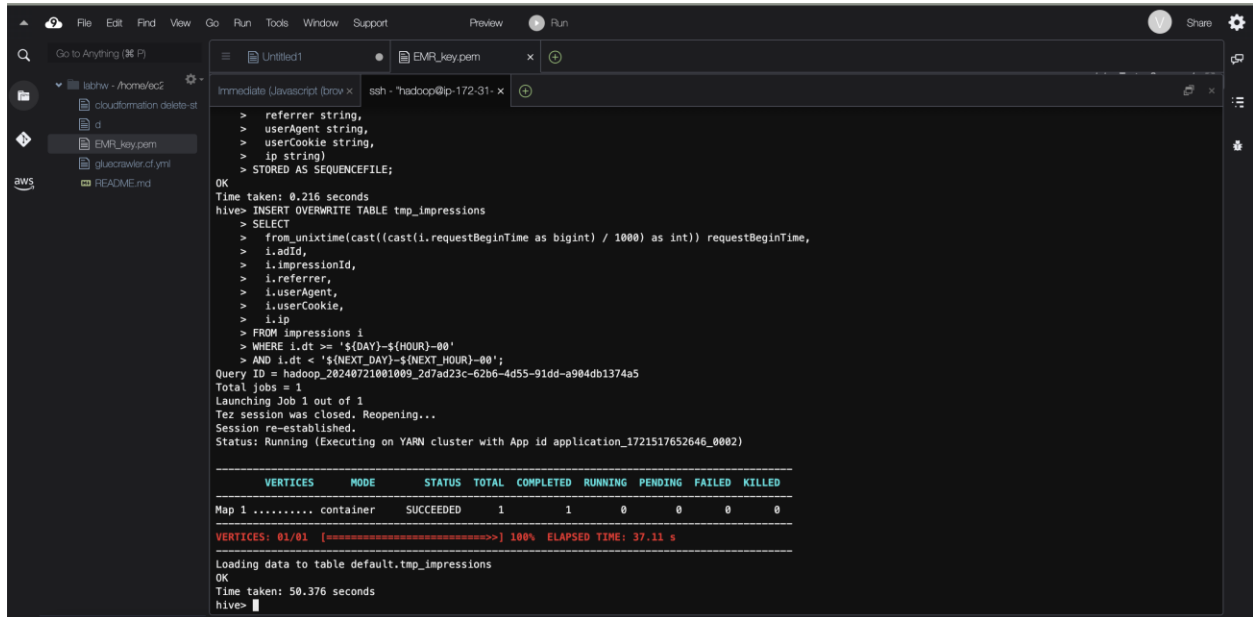
Task 5: Using Hive to Join Tables

In this work, you will create a new table called joined impressions by joining the impressions and clicks tables. This gives you insights into user behavior by combining data about impression information about advertising that users saw with clicks information about ads that users clicked on.

```
SerDe Library:         org.apache.hive.hcatalog.data.JsonSerDe
InputFormat:           org.apache.hadoop.mapred.TextInputFormat
OutputFormat:          org.apache.hadoop.hive ql.io.HiveIgnoreKeyTextOutputFormat
Compressed:            No
Num Buckets:           -1
Bucket Columns:        []
Sort Columns:          []
Storage Desc Params:
  paths                impressionId
  serialization.format  1
Time taken: 0.114 seconds, Fetched: 38 row(s)
hive> CREATE EXTERNAL TABLE joined_impressions (
  > requestBeginTime string,
  > adid string,
  > impressionid string,
  > referrer string,
  > userAgent string,
  > userCookie string,
  > ip string,
  > clicked Boolean)
  > PARTITIONED BY (day string, hour string)
  > STORED AS SEQUENCEFILE
  > LOCATION '${OUTPUT}/tables/joined_impressions';
OK
Time taken: 1.277 seconds
hive> CREATE TABLE tap_impressions (
  > requestBeginTime string,
  > adid string,
  > impressionid string,
  > referrer string,
  > userAgent string,
  > userCookie string,
  > ip string)
  > STORED AS SEQUENCEFILE;
OK
Time taken: 0.216 seconds
hive>
```

Now, we will be creating a joined impressions table. This table will have the same columns as the impressions and one more column which is clicked to indicate if the ad is clicked or not.

Creating a table for temporary impressions to store the data.



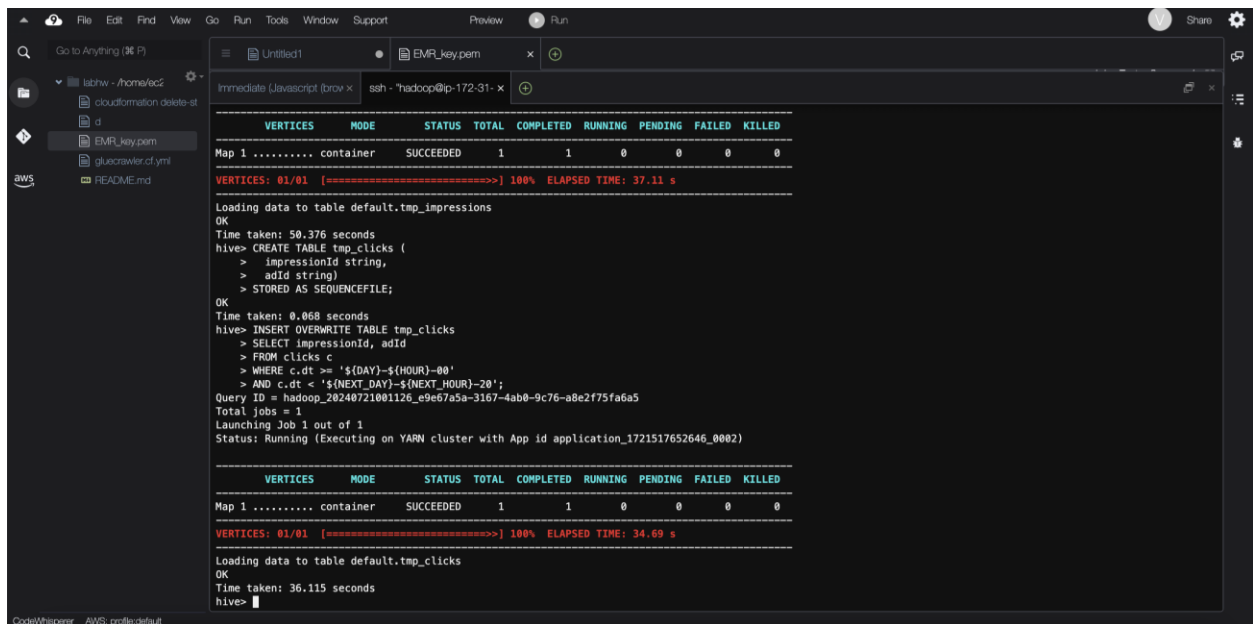
```
File Edit Find View Go Run Tools Window Support Preview Run Share
Go to Anything (⌘ P)
labhw - /home/ec2
cloudformation delete-st
d
EMR_key.pem
gluecrawler.cf.yml
README.md
aws

Immediate Javascript (brow x) ssh - hadoop@ip-172-31-...
> referrer string,
> userAgent string,
> userCookie string,
> ip string)
> STORED AS SEQUENCEFILE;
OK
Time taken: 0.216 seconds
hive> INSERT OVERWRITE TABLE tmp_impressions
> SELECT
>   from_unixtime(cast((cast(i.requestBeginTime as bigint) / 1000) as int)) requestBeginTime,
>   i.adId,
>   i.impressionId,
>   i.referrer,
>   i.userAgent,
>   i.userCookie,
>   i.ip
> FROM impressions i
> WHERE i.dt >= '${DAY}-${HOUR}-00'
> AND i.dt < '${NEXT_DAY}-${NEXT_HOUR}-00';
Query ID = hadoop_20240721001009_2d7ad23c-62b6-4d55-91dd-a984db1374a5
Total jobs = 1
Launching Job 1 out of 1
Tez session was closed. Reopening...
Session re-established.
Status: Running (Executing on YARN cluster with App id application_1721517652646_0002)

VERTICES   MODE        STATUS TOTAL COMPLETED RUNNING PENDING FAILED KILLED
Map 1 ..... container  SUCCEEDED 1      1      0      0      0      0
VERTICES: 01/01 [=====] 100% ELAPSED TIME: 37.11 s

Loading data to table default.tmp_impressions
OK
Time taken: 50.376 seconds
hive>
```

Inserting data into temp_impressions table for certain time duration. And the time took for this is 50.37 seconds.



```
File Edit Find View Go Run Tools Window Support Preview Run Share
Go to Anything (⌘ P)
labhw - /home/ec2
cloudformation delete-st
d
EMR_key.pem
gluecrawler.cf.yml
README.md
aws

Immediate Javascript (brow x) ssh - hadoop@ip-172-31-...
VERTICES   MODE        STATUS TOTAL COMPLETED RUNNING PENDING FAILED KILLED
Map 1 ..... container  SUCCEEDED 1      1      0      0      0      0
VERTICES: 01/01 [=====] 100% ELAPSED TIME: 37.11 s

Loading data to table default.tmp_impressions
OK
Time taken: 50.376 seconds
hive> CREATE TABLE tmp_clicks (
>   impressionId string,
>   adId string)
> STORED AS SEQUENCEFILE;
OK
Time taken: 0.068 seconds
hive> INSERT OVERWRITE TABLE tmp_clicks
> SELECT impressionId, adId
> FROM clicks c
> WHERE c.dt >= '${DAY}-${HOUR}-00'
> AND c.dt < '${NEXT_DAY}-${NEXT_HOUR}-20';
Query ID = hadoop_20240721001126_e9e67a5a-3167-4ab0-9c76-a8e2f75fa6a5
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1721517652646_0002)

VERTICES   MODE        STATUS TOTAL COMPLETED RUNNING PENDING FAILED KILLED
Map 1 ..... container  SUCCEEDED 1      1      0      0      0      0
VERTICES: 01/01 [=====] 100% ELAPSED TIME: 34.69 s

Loading data to table default.tmp_clicks
OK
Time taken: 36.115 seconds
hive>
```

Now, creating one more table for temporary clicks for storing the click data.

```
File Edit Find View Go Run Tools Window Support Preview Run Share
Go to Anything (⌘ P)
lathw - /home/ec2
cloudformation delete-st
d
EMR_key.pem
gluecrawler.cf.yml
README.md
aws

Untitled1
EMR_key.pem
ssh - hadoop@ip-172-31-...
Immediate Javascript (browser)
VERTICES: 01/01 [=====] 100% ELAPSED TIME: 34.69 s
Loading data to table default.tmp_clicks
OK
Time taken: 36.115 seconds
hive> INSERT OVERWRITE TABLE joined_impressions
> PARTITION (day='${DAY}', hour='${HOUR}')
> SELECT
>   i.requestBeginTime,
>   i.adId,
>   i.impressionId,
>   i.referrer,
>   i.userAgent,
>   i.userCookie,
>   i.ip,
>   (c.impressionId is not null) clicked
> FROM tmp_impressions i
> LEFT OUTER JOIN tmp_clicks c ON i.impressionId=c.impressionId;
Query ID = hadoop_20240721001234_708cde14-d8a0-477a-b150-d3acece3da8b
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1721517652646_0002)

VERTICES   MODE   STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
Map 1 ..... container  SUCCEEDED    1         1         0         0         0         0
Map 2 ..... container  SUCCEEDED    1         1         0         0         0         0
VERTICES: 02/02 [=====] 100% ELAPSED TIME: 15.77 s
Loading data to table default.joined_impressions partition (day=2009-04-13, hour=08)
OK
Time taken: 19.7 seconds
hive> set hive.cli.print.header=true;
hive>
```

Now inserting the data into the temp_clicks table for slightly longer duration to see its performance.

```
File Edit Find View Go Run Tools Window Support Preview Run Share
Go to Anything (⌘ P)
lathw - /home/ec2
cloudformation delete-st
d
EMR_key.pem
gluecrawler.cf.yml
README.md
aws

Untitled1
EMR_key.pem
ssh - hadoop@ip-172-31-...
Immediate Javascript (browser)
VERTICES   MODE   STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
Map 1 ..... container  SUCCEEDED    1         1         0         0         0         0
Map 2 ..... container  SUCCEEDED    1         1         0         0         0         0
VERTICES: 02/02 [=====] 100% ELAPSED TIME: 15.77 s
Loading data to table default.joined_impressions partition (day=2009-04-13, hour=08)
OK
Time taken: 19.7 seconds
hive> set hive.cli.print.header=true;
hive> SELECT * FROM joined_impressions LIMIT 10;
OK
joined_impressions.requestbeginTime  joined_impressions.adid  joined_impressions.impressionid  joined_impressions.referrer  joined_impressions.userAgent  joi
ned_impressions.usercookie  joined_impressions.ip  joined_impressions.clicked  joined_impressions.day  joined_impressions.hour
2009-04-13 08:02:27  HQMHYITLdSUSWnJiGMLMPdW79102B  FLwMbIqW1stIXdBPuDJCH43FSC0bq  cartoonnetwork.com  Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 6.0;
FunWebProducts; GTB6; SLCC1; .NET CLR 2.0.50727; Media Center PC 5.0; .NET wcVMTascoPbGt6bdqDbuWTPPhq0Ps 69.191.224.234 false 2009-04-13 08
2009-04-13 08:10:10  wM8Tfrdq1HoQX41TMH0wSLuJqLBNdw  FSLolrpIngS8qKP5TtPHt30AF4eG6  cartoonnetwork.com  Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 6.0;
FunWebProducts; GTB6; SLCC1; .NET CLR 2.0.50727; Media Center PC 5.0; .NET wcVMTascoPbGt6bdqDbuWTPPhq0Ps 69.191.224.234 false 2009-04-13 08
2009-04-13 08:02:03  pu6cmp9VAAT6Fu48JfrtuVL3b9Hqwe  lgW0gftxXiaedGCMhsVG6BpG3FFGhp  cartoonnetwork.com  Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.2;
SV1; .NET CLR 1.1.4322; InfoPath.1)  Q5b3wKLR4JA1ut4Uq6FNFIr1rCvWu  42.174.193.253  false 2009-04-13 08
2009-04-13 08:04:16  J1rppKBUvpk)JT1eHSAXD0GAVQ910K  BNGIPAG6VU02A2T1ue2B3DWMHUEgg  cartoonnetwork.com  Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.2;
SV1; .NET CLR 1.1.4322; InfoPath.1)  Q5b3wKLR4JA1ut4Uq6FNFIr1rCvWu  42.174.193.253  false 2009-04-13 08
2009-04-13 08:05:25  AgJUsIKd6F2KvT9eBRA6PGSVVLOP7A  hCc0NweliHw8RDKsL49qb)l38TE83u  cartoonnetwork.com  Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.2;
SV1; .NET CLR 1.1.4322; InfoPath.1)  Q5b3wKLR4JA1ut4Uq6FNFIr1rCvWu  42.174.193.253  false 2009-04-13 08
2009-04-13 08:06:35  VwTcF8G30A700qf0gBB173K6BKGf5  uTX1FkBsImckMcnacu3XpGLKJ)MR3IC  cartoonnetwork.com  Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.2;
SV1; .NET CLR 1.1.4322; InfoPath.1)  Q5b3wKLR4JA1ut4Uq6FNFIr1rCvWu  42.174.193.253  false 2009-04-13 08
2009-04-13 08:01:11  0UUBqMx3X133mRn0IKBksS5Uj07Toe  S01Ks0aJs54iMHEkm7Kgmduw4UR6  cartoonnetwork.com  Mozilla/4.0 (compatible; MSIE 8.0; Windows NT 5.1;
Trident/4.0; GTB6; .NET CLR 1.1.4322)  K96nSPnuumwMKfIU10TFP0TnMfADq9  51.131.29.87  false 2009-04-13 08
2009-04-13 08:03:26  4FC0Kqg40f10K8B6f8mcqAg5qVg6  p250GfBuxT0HkqWqHqN10HcV2Lc  cartoonnetwork.com  Mozilla/4.0 (compatible; MSIE 8.0; Windows NT 5.1;
Trident/4.0; GTB6; .NET CLR 1.1.4322)  K96nSPnuumwMKfIU10TFP0TnMfADq9  51.131.29.87  false 2009-04-13 08
2009-04-13 08:07:48  wraqhTKLXsVnVq9k43GmBqsn434FD  GurRl13BUETwTvtVw0XcsiHPaKnrI0  cartoonnetwork.com  Mozilla/4.0 (compatible; MSIE 8.0; Windows NT 5.1;
Trident/4.0; GTB6; .NET CLR 1.1.4322)  K96nSPnuumwMKfIU10TFP0TnMfADq9  51.131.29.87  false 2009-04-13 08
2009-04-13 08:02:43  FUQXd0aav8FrEhFpWw6qdA1H16PP7  RJ5RFN4r09kkuCXE4E1f9chQ08tnha  cartoonnetwork.com  Mozilla/5.0 (Macintosh; U; Intel Mac OS X 10.5; en-
US; rv:1.9.1.1) Gecko/20090715 Firefox/3.5.1  ewDEVUvphLnRa273j3jBSlMG6weW7m  22.91.173.232  false 2009-04-13 08
Time taken: 0.194 seconds, Fetched: 10 row(s)
hive>
```

Performing the join of temp_impressions and temp_clicks table and writing the results into joined_impressions table.

Using the hive commands, printing the column names as header.


```
File Edit Find View Go Run Tools Window Support Preview Run Share
Go to Anything (⌘ P)
labhw - /home/lab2
cloudformation delete-st
d
EMR_key.pem
gluecrawler.cf.yml
README.md
aws
Untitled1
EMR_key.pem
Immediate (Javascript) (brow x) ssh - "hadoop@ip-172-31-1-
2009-04-13 08:02:43 FUQX0aav8FrEhFpWw6qdA1H16PP7 RJ5RfM4rQ9kkuCXE4EIf9chQ88taha cartoonnetwork.com Mozilla/5.0 (Macintosh; U; Intel Mac OS X 10.5; en-US; rv:1.9.1.1) Gecko/20090715 Firefox/3.5.1 ewDEVVuphlnRa273j3b5UMG6weW7m 22.91.173.232 false 2009-04-13 08
Time taken: 0.194 seconds, Fetched: 10 row(s)
hive> SELECT adid, count(*) AS hits
> FROM joined_impressions
> WHERE clicked = true
> GROUP BY adid
> ORDER BY hits DESC
> LIMIT 10;
Query ID = hadoop_20240721001335_7a53c9f7-da1b-42d8-aa47-ae9cca098c27
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1721517652646_0002)

VERTICES   MODE   STATUS   TOTAL   COMPLETED   RUNNING   PENDING   FAILED   KILLED
Map 1 ..... container   SUCCEEDED   1       1       0       0       0       0
Reducer 2 ..... container   SUCCEEDED   1       1       0       0       0       0
Reducer 3 ..... container   SUCCEEDED   1       1       0       0       0       0
VERTICES: 03/03 [=====] 100% ELAPSED TIME: 11.07 s
OK
adid hits
70n4fCvgAV0wfojpw5sw30AEiaUUMg 5
FxSntm0CqAx6LejJR2qVelsXoFCXL7 4
2IkjC5AbFI6IA3LGrhATQgpd07KhAk 4
7p32sreTnFKIK5gLSkCmexjX1XL8P 4
HdQwXn703hQUJINoSS4JumGN79w 4
tbKLLXhArCiqQMj3fII07rpGuldIKJ 3
90B0cF2ejuljyPHBfBwvdl4S41cl6 3
8NjKlm78sc6i9cc1NvHLT3v6Ua4N2 3
vgA0hCFRsbw4CFnwGJoaTul1eIqFQc 3
5C1fCHhj8mk2js85ERhCLOfCuriEdN 3
Time taken: 12.641 seconds, Fetched: 10 row(s)
hive>
```

To find the 10 most clicked ads IDs and to get the first 10 rows of from the joined_impressions table we will be using the above commands.

```
File Edit Find View Go Run Tools Window Support Preview Run Share
Go to Anything (⌘ P)
labhw - /home/lab2
cloudformation delete-st
d
EMR_key.pem
gluecrawler.cf.yml
README.md
aws
Untitled1
EMR_key.pem
Immediate (Javascript) (brow x) ssh - "hadoop@ip-172-31-1-
8NjKlm78sc6i9cc1NvHLT3v6Ua4N2 3
vgA0hCFRsbw4CFnwGJoaTul1eIqFQc 3
5C1fCHhj8mk2js85ERhCLOfCuriEdN 3
Time taken: 12.641 seconds, Fetched: 10 row(s)
hive> exit;
[hadoop@ip-172-31-35-51 ~]$ hive -e "SELECT referrer, count(*) as hits FROM joined_impressions WHERE clicked = true GROUP BY referrer ORDER BY hits DESC LIMIT 10;"
> /home/hadoop/result.txt

Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j2.properties Async: true
Query ID = hadoop_20240721001434_072479ba-456d-41b1-a7be-3e1f57ede764
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1721517652646_0004)

Map 1: ~/-/ Reducer 2: 0/1 Reducer 3: 0/1
Map 1: 0/1 Reducer 2: 0/1 Reducer 3: 0/1
Map 1: 0/1 Reducer 2: 0/1 Reducer 3: 0/1
Map 1: 0(+1)/1 Reducer 2: 0/1 Reducer 3: 0/1
Map 1: 0(+1)/1 Reducer 2: 0/1 Reducer 3: 0/1
Map 1: 1/1 Reducer 2: 0/1 Reducer 3: 0/1
Map 1: 1/1 Reducer 2: 0(+1)/1 Reducer 3: 0/1
Map 1: 1/1 Reducer 2: 1/1 Reducer 3: 0(+1)/1
Map 1: 1/1 Reducer 2: 1/1 Reducer 3: 1/1
OK
Time taken: 29.369 seconds, Fetched: 10 row(s)
[hadoop@ip-172-31-35-51 ~]$ cat /home/hadoop/result.txt
ibibo.com 49
lowes.com 33
odnoklasesniki.ru 33
files.wordpress.com 30
media-servers.net 29
tmz.com 29
tattoodile.com 29
tiscali.it 28
mpnrs.com 28
barnesandnoble.com 28
[hadoop@ip-172-31-35-51 ~]$
```

Now we will have to exit from the hive and save all that data into s3 bucket. We must find out the path and using cat commands we will be saving those results into that bucket at specified locations.

Conclusion:

In this laboratory we implemented the entire process of data warehousing with AWS EMR-HIVE machinery from padding of bare necessities needed to handle the infrastructure and finally using SQL-like queries to analyze the data.

At first, launching an EMR cluster was required by the laboratory which is important for processing large scale data. Through AWS Cloud9, you SSHed into the EMR cluster and configured a strong environment that supported all future activities.

These tables, which were stored in json format, were very relevant when it came to understanding user interaction with ads. Thus designing these two tables and telling where they will be located on S3's gives you foundation of good places for storage retrieval.

To elevate the performance of the datasets even higher, you turned these datasets into external tables with SequenceFile format. This transformation was made using HiveQL commands that took advantage of Hadoop's built-in compression features leading to great improvements in processing speediness.

The above move saw the creation of tables namely impressions_seq and clicks_seq which enables smooth data accessing thereby lowering space requirements. In keeping with the basic structure, your focus shifted to another essential duty of merging impressions data with clicks of information.

The aim was to combine such sources into a unified region where one could understand whose advertisements were clicked by users. To achieve this end goal, you set up a fresh external table known as joined_impressions intended to store all collected figures derived from both sources. An extra column was added to this table showing if an impression led to a click or not.

After you filled in the table called "joined impressions," you used Hive to query the data. While configuring Hive CLI to include column names in its output display, you wrote some queries that extracted and examined the data. The earliest return offered an initial short summary while the last ones revealed which ads were clicked on most frequently as well as from which sites or sources, this shed light on ad performance and how much users interacted with them.

You have shown during this lab that it is possible to handle big volume datasets, optimize data storage and perform complicated transformations and analysis using Hive or any other application running on EMR of Amazon.

By making sure that you put the resulting data into S3, you have ensured that they would always be there for future reference even when an EMR cluster is no longer running. Such an approach thus meets the expensive processing needs regarding data and cheap storage alternatives.